



Section 1

Simple Storage Service (S3)

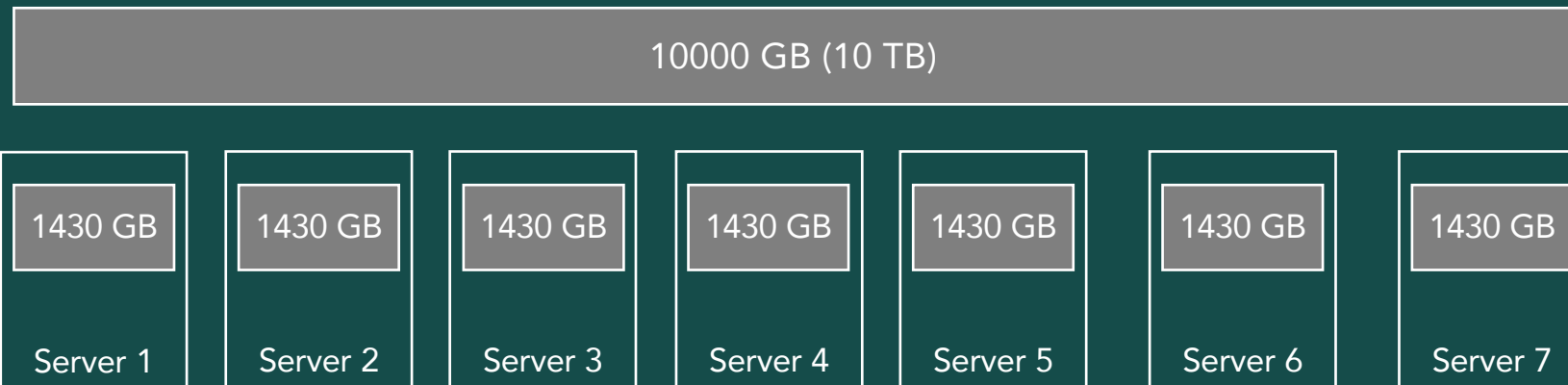
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Agenda

- Distributed Storage
- S3 Object Store
- S3 in Data Engineering Pipeline
- Use Cases of S3
- Block Store and Object Store

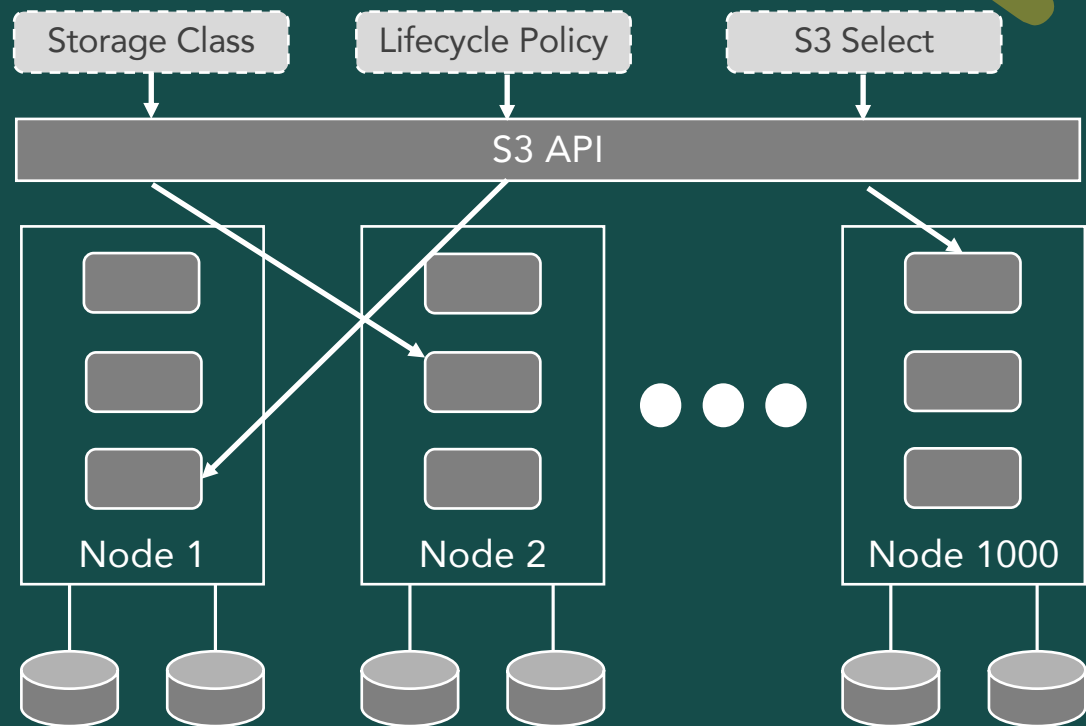
Distributed Storage

- Limitation on a single server.
- Break the large file in small blocks and store them in multiple servers.
- Will only keeping the blocks help?



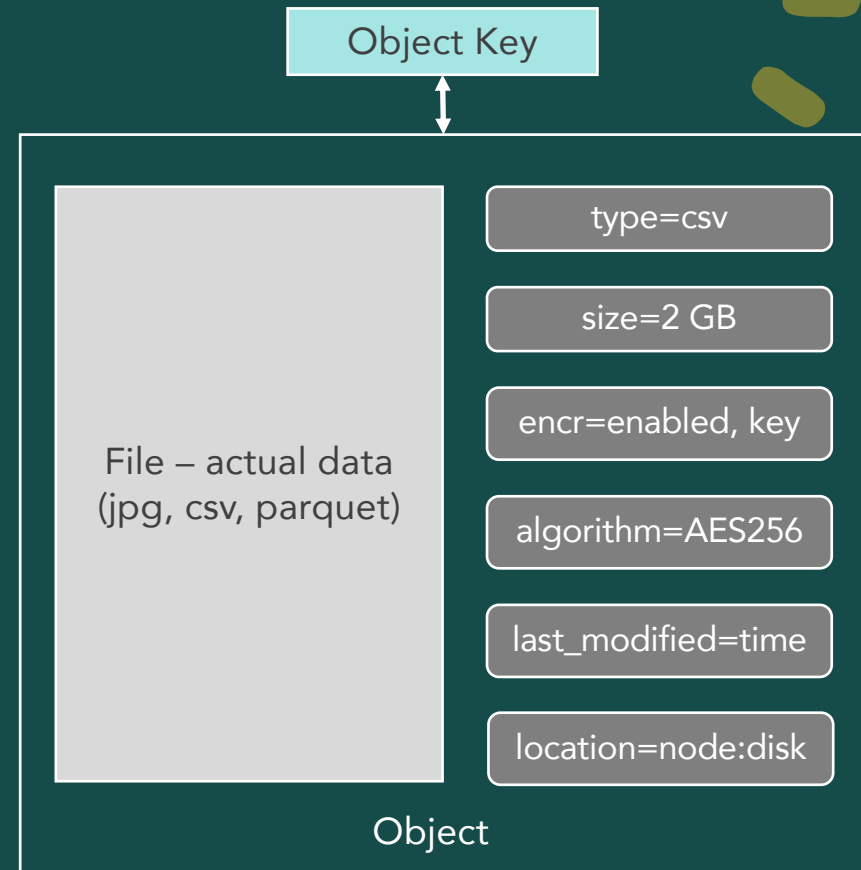
Simple Storage Service (S3)

- Distributed storage. But, its not only that!!
- Multiple services / programs run in the nodes.
- These services help achieve multiple functionalities and features.
- S3 API is given to end users.

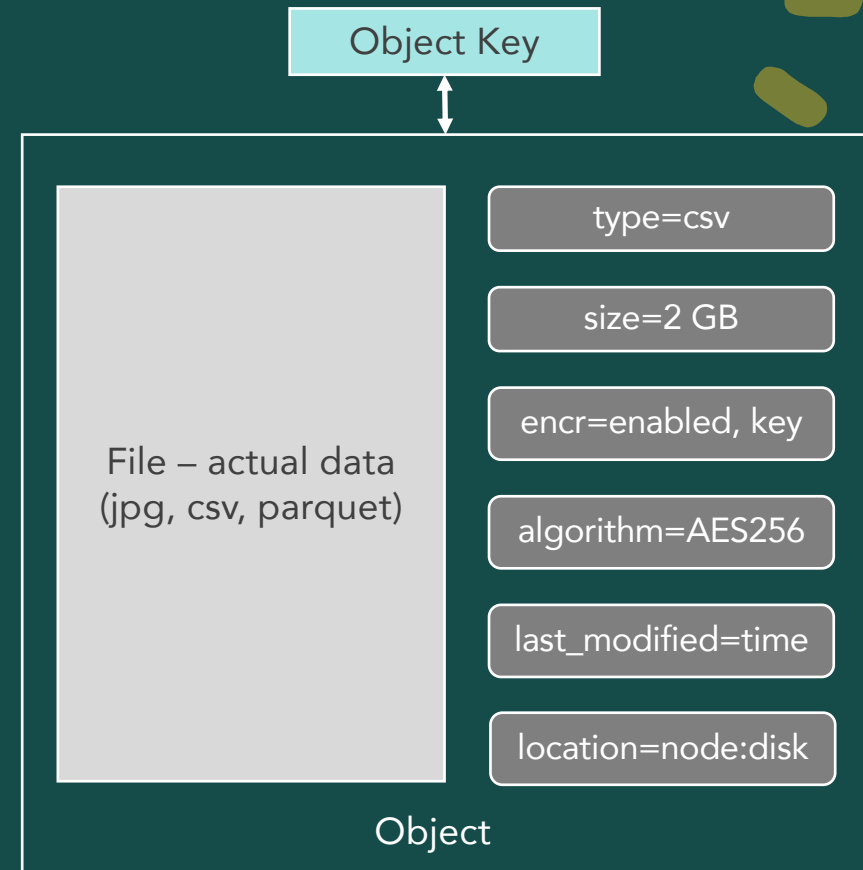
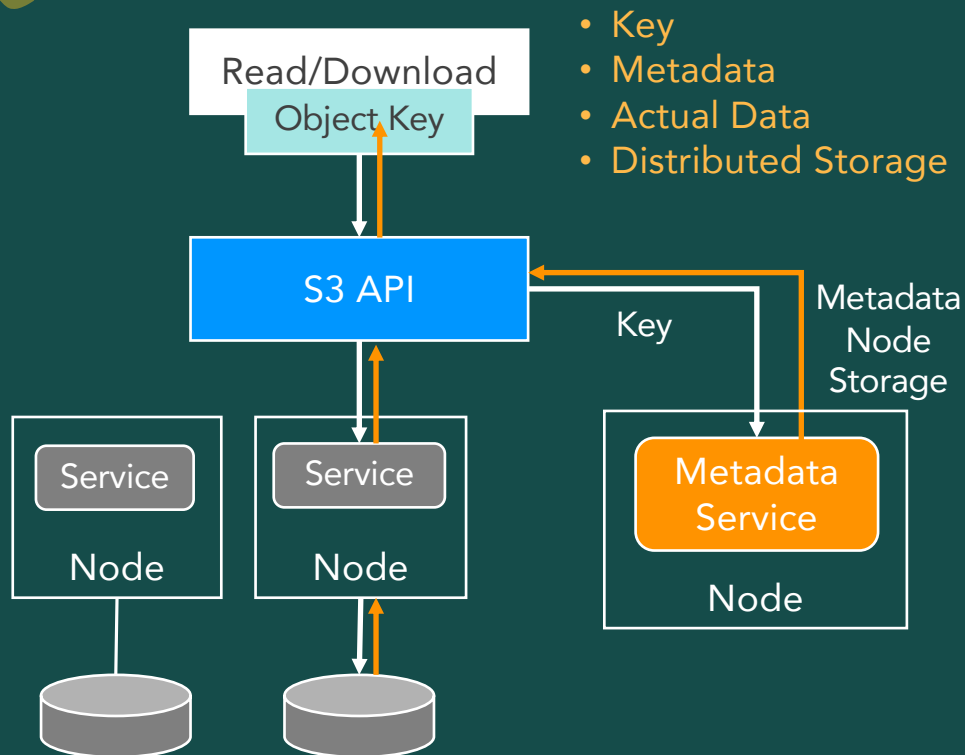


S3 Object Store

- Any kind of data.
 - Files – CSV, Parquet, Avro, JSON, ORC etc.
 - Images – jpg, png, jpeg etc.
 - Videos – mp4, mov etc.
- Stores metadata of the file as well.
 - Type of file.
 - Size of file.
 - Last modified time.
 - Encryption.
 - Tags.

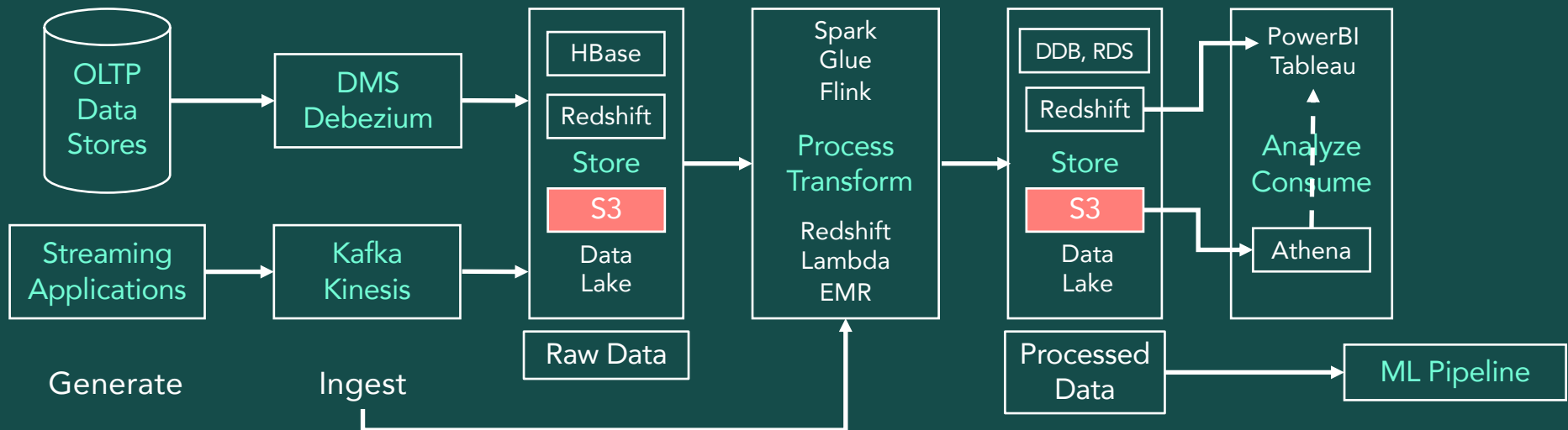


S3 Object Store



S3 in Data Engg. Pipeline

- Data sources - Generate.
- Ingest & Collect. Store.
- Process & Transform. Store.
- Analyze & Consume.

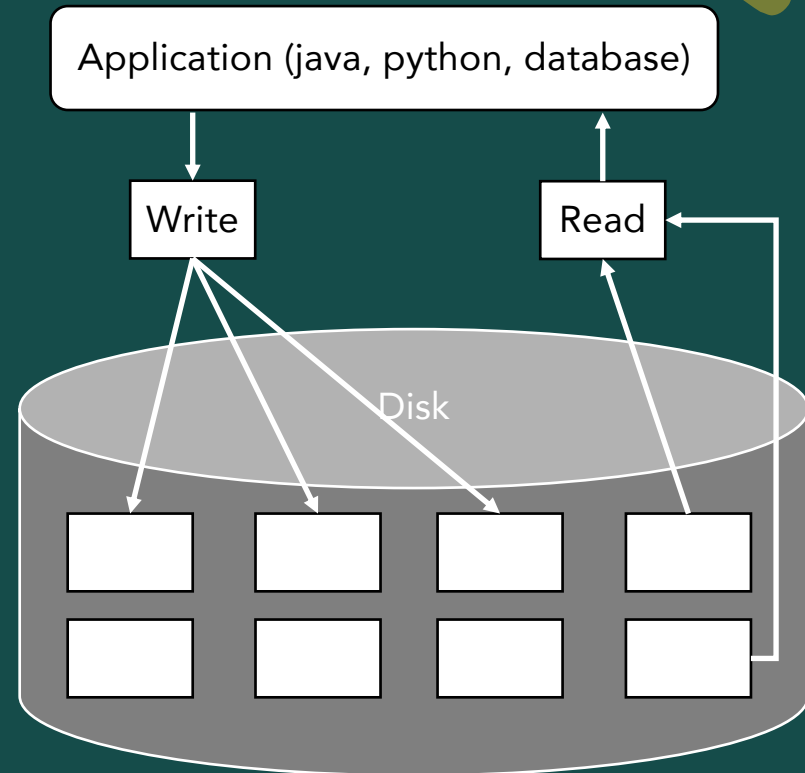


Use Cases of S3

- Data Lake – Raw data & Processed data.
- Lakehouse – Iceberg, HUDI, Delta Lake.
- Distributed Storage.
- Data Archival.
- Backup Storage.
- Application log files.
- Media content.

Object Store & Block Store

- Data is stored in Blocks. EBS.
- Data is read and written in blocks.
- Can we create object store using block store?
 - YES!
- Problems??
 - Management, Scaling, Availability.





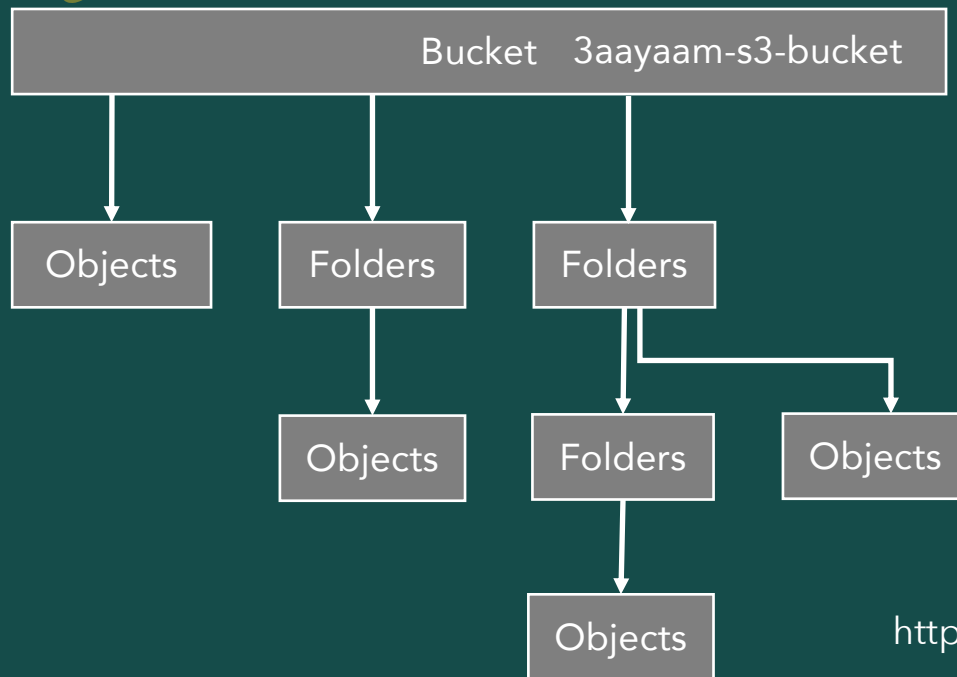
S3 Basics

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Agenda

- S3 Storage Hierarchy
- Buckets & Objects
- Bucket Properties – versioning, encryption, server access logging, event notifications, transfer acceleration, requestor pays.
- Object properties – storage class, multipart upload, object operations, object locks.

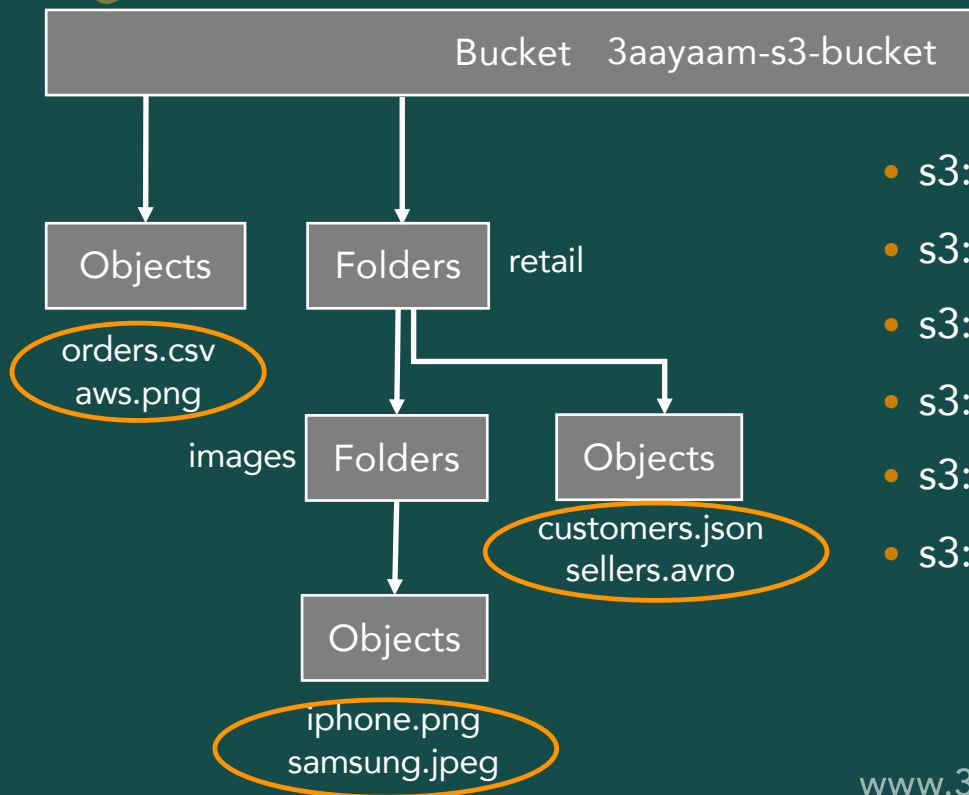
Storage Hierarchy



- Bucket
- Objects – csv, parquet, avro, json, jpeg, png, mp4 etc.
- Folders
- URI & URL
 - URI – `s3://3aayaam-s3-bucket/`
 - URL – `https://3aayaam-s3-bucket.s3.<region-code>.amazonaws.com/`

`https://3aayaam-s3-bucket.s3.us-east-1.amazonaws.com/`

Access Buckets & Objects

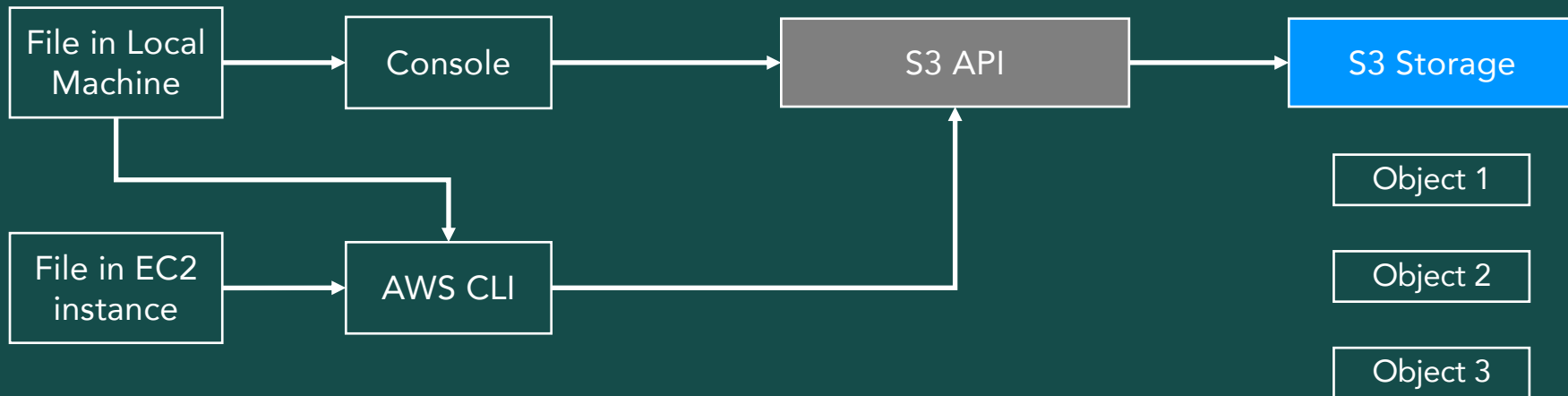


- `s3://3aayaam-s3-bucket/orders.csv`
- `s3://3aayaam-s3-bucket/aws.png`
- `s3://3aayaam-s3-bucket/retail/customers.json`
- `s3://3aayaam-s3-bucket/retail/sellers.avro`
- `s3://3aayaam-s3-bucket/retail/images/iphones.png`
- `s3://3aayaam-s3-bucket/retail/images/samsung.jpeg`

Object Keys

How to upload and store Objects

- AWS Console.
- AWS CLI.



Hands-on

- Create bucket.
- Create folder.
- Upload objects.
 - AWS Console.
 - AWS CLI from EC2 instance.

Common S3 AWS CLI commands

- `aws s3 ls`
- `aws s3 cp`
- `aws s3 mv`
- `aws s3 rm`
- `aws s3api`

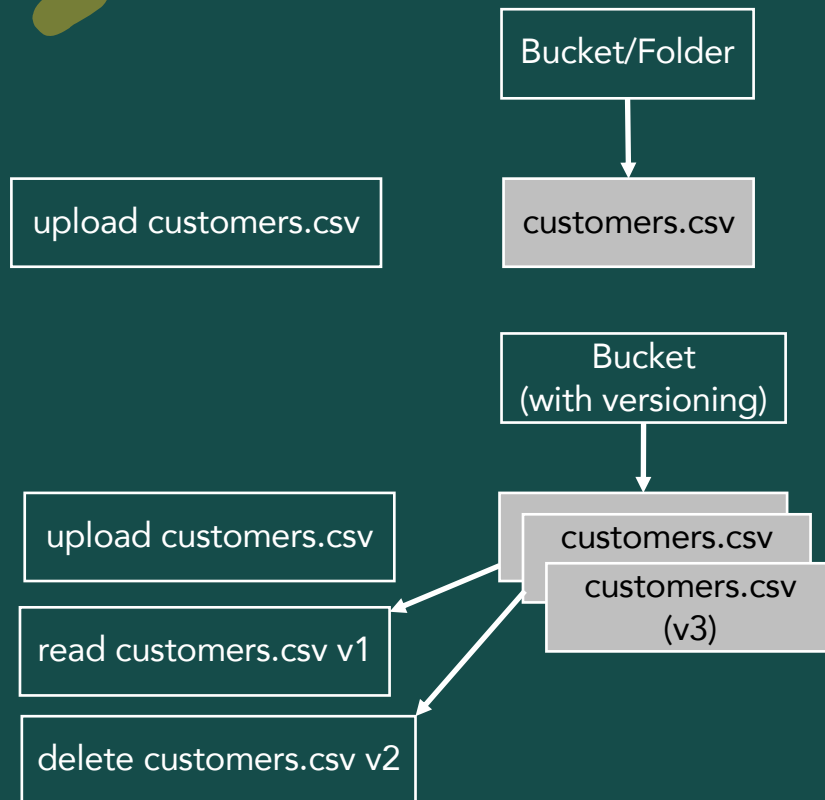


Bucket Properties



- Versioning
- Encryption
- Server Access Logging
- Event Notifications
- Transfer Acceleration
- Requestor Pays
- Object Locking

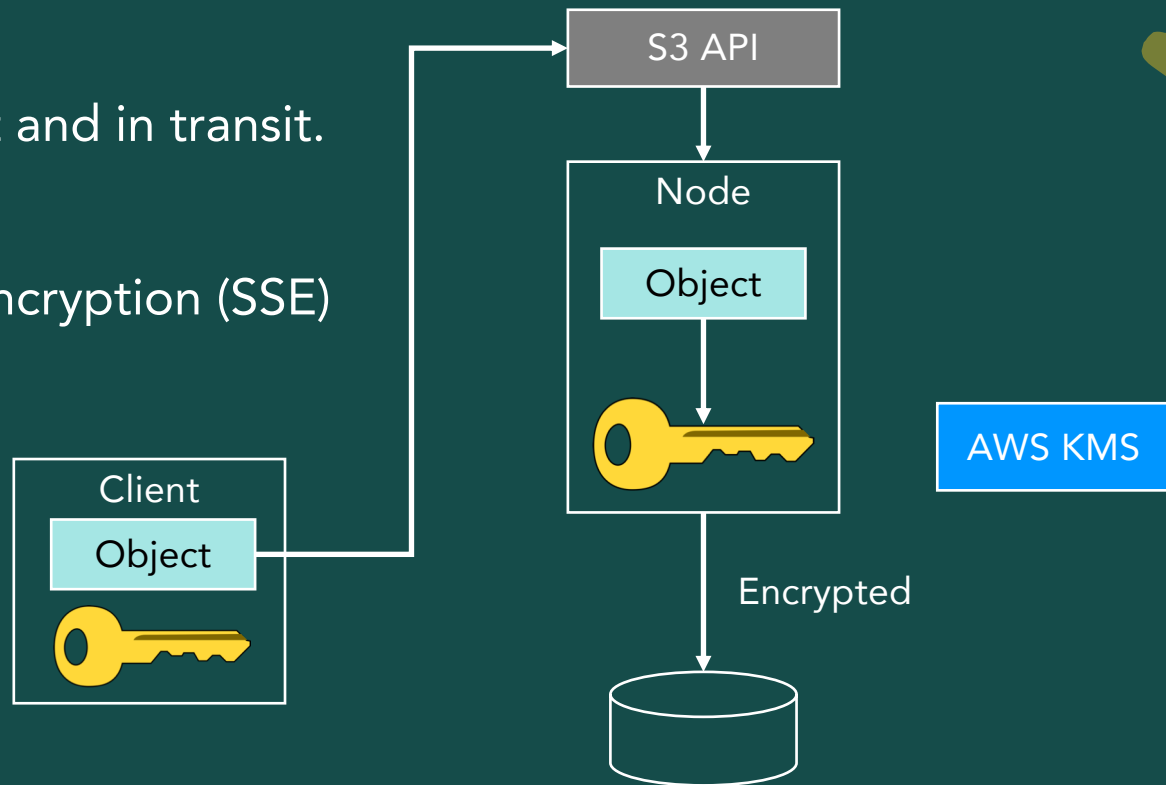
Versioning



- Multiple versions of the same object are kept.
- Helps against accidental upload, delete etc. Historical purposes.
- Version IDs are automatically generated by S3.
- Console & CLI – aws s3api
 - `aws s3api list-object-versions --bucket <bucket>`
 - `aws s3api delete-object --bucket <b1> --key <object-key> --version-id <id>`

Encryption

- Data encryption at rest and in transit.
- In transit – SSL/TLS
- At rest – Server Side Encryption (SSE)
 - SSE-S3
 - SSE-KMS
 - DSSE-KMS
 - SSE-C





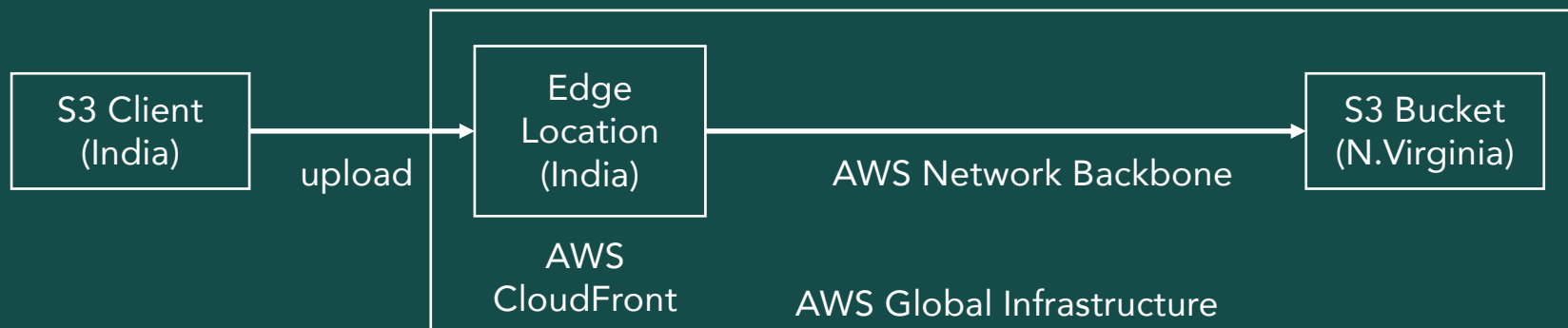
Server Access Logging



- Log all access to S3.
- Keeps detailed information about all requests that are made to a bucket.
- Security & audit.

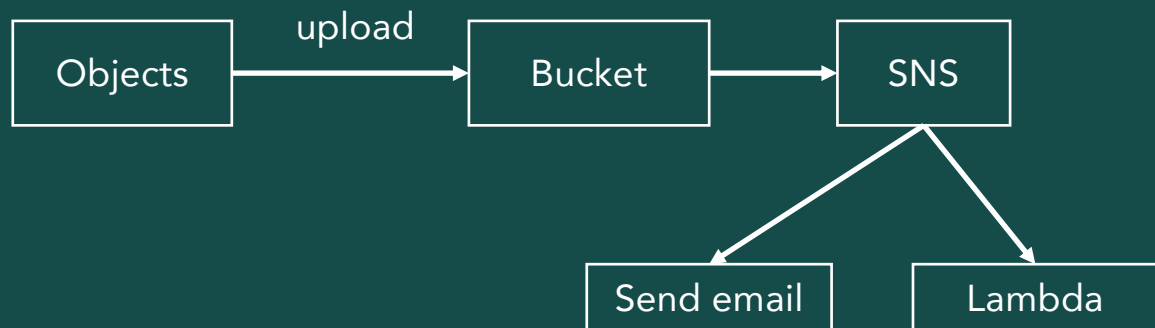
Transfer Acceleration & Requestor Pays

- Fast transfer of objects from clients to S3 over long distances.
- Different bucket endpoint has to be used.
- Additional price.
- Requestor Pays – Cost of request and download are owned by the requestor.



Event Notifications

- Notifications based on events in S3 bucket.
- Events:
 - Upload objects.
 - Remove objects.
 - Replication.
 - Etc.





S3 Object Locks



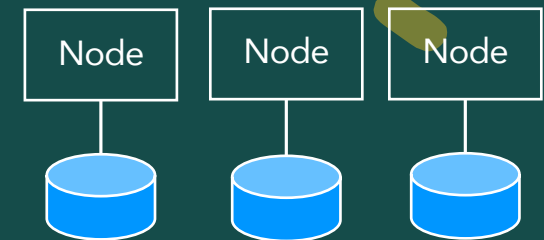
- Restricts objects to get deleted or overwritten.
- Has to be enabled at Bucket level.
- WORM (write-once-read-many) model.
- Type of object locks:
 - Retention period – Duration for which the object cannot be modified.
 - Legal hold – Unlimited retention. Cannot be deleted unless legal hold is removed.
 - Configured at object level.

Hands-on

- Done

S3 Storage Class

- Provides different storage types, access latency and availability based on usage. **Mentioned at object level.**
- S3 Standard
 - High durability, availability, and performance for **frequently accessed data.**
 - Low latency and high throughput.
 - Stored in at least 3 AZs.
 - Use case: Analytics.
- S3 Standard-IA
 - High durability, throughput, low latency. For **less frequently accessed data.**
 - Stored in at least 3 AZs.
 - Use case: DR, backups, higher duration for compliance, log storage.



S3 Storage Class

- S3 Express One Zone
 - High performance, single AZ. For frequently accessed data.
 - Low latency single-digit millisecond data access. 10x faster than S3 Standard.
 - Better to store small sized objects.
 - Use case: Analytics, ML.
- S3 One Zone-IA
 - Same performance and latency of S3 Standard.
 - Less frequently accessed data.

S3 Storage Class

- Archive storage class. To archive data.
- S3 Glacier Instant Retrieval
 - Archive data. Retrieve once per quarter. Immediate retrieval.
 - Millisecond access latency as Standard and Standard-IA storage classes.
 - Medical records, media content, documents.
 - Can be uploaded directly or using Lifecycle Policies.
- S3 Glacier Flexible Retrieval
 - Access requirement once or twice a year. Asynchronous retrieval with minutes to hours latency.
 - Less frequently accessed data.
 - Backup, DR.

S3 Storage Class

- S3 Glacier Deep Archive

- Long-term preservation.
- Compliance requirement to store data for 5-15 years.
- Retrieval time is within 12 hrs.
- Magnetic tape archives.

- S3 Intelligent Tiering

- Automatically moves objects across storage classes based on access frequency.

- Pricing

- Standard – \$0.023/GB/month. Standard-IA - \$0.0125/GB/month.
- Express One Zone - \$0.16/GB/month. One Zone-IA - \$0.01/GB/month.
- Glacier Instance Retrieval - \$0.004/GB/month. Glacier Flexible Retrieval - \$0.0036/GB/month. Glacier Deep Archive - \$0.00099/GB/month.

S3 Storage Class

Storage class	Designed for	Durability (designed for)	Availability (designed for)	Availability Zones	Min storage duration
S3 Standard	Frequently accessed data (more than once a month) with millisecond access	99.99999999%	99.99%	>= 3	None
S3 Standard-IA	Long-lived, infrequently accessed data (once a month) with millisecond access	99.99999999%	99.9%	>= 3	30 days
S3 Intelligent-Tiering	Data with unknown, changing, or unpredictable access patterns	99.99999999%	99.9%	>= 3	None
S3 One Zone-IA	Recreatable, infrequently accessed data (once a month) with millisecond access	99.99999999%	99.5%	1	30 days
S3 Express One Zone	Single-digit millisecond data access for latency-sensitive applications within a single AWS Availability Zone	99.99999999%	99.95%	1	None

S3 Storage Class

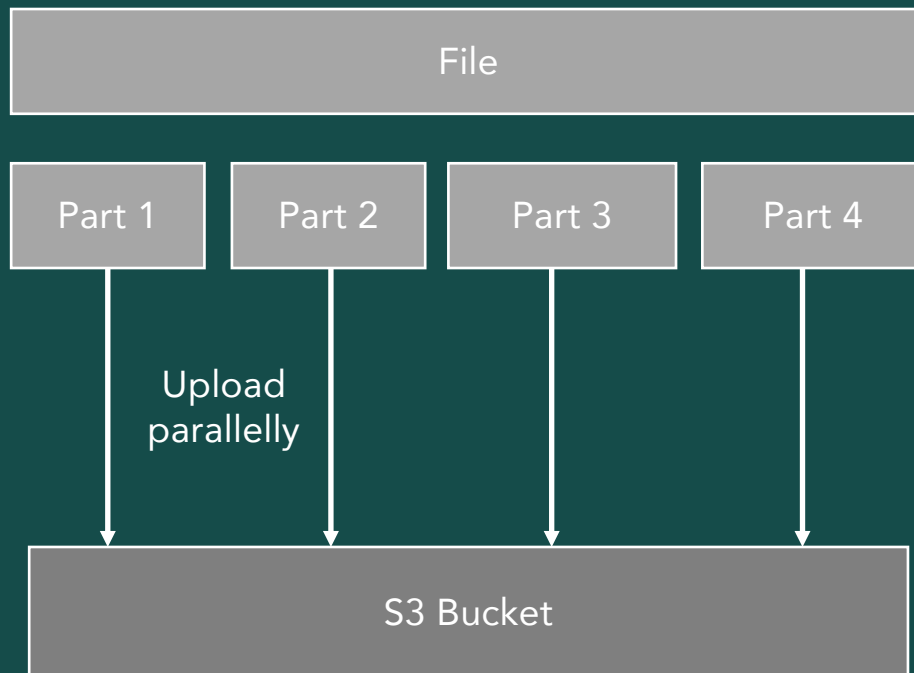
S3 Glacier Instant Retrieval	Long-lived, archive data accessed once a quarter with millisecond access	99.999999999%	99.9%	≥ 3	90 days
S3 Glacier Flexible Retrieval	Long-lived archive data accessed once a year with retrieval times of minutes to hours	99.999999999%	99.99% (after you restore objects)	≥ 3	90 days
S3 Glacier Deep Archive	Long-lived archive data accessed less than once a year with retrieval times of hours	99.999999999%	99.99% (after you restore objects)	≥ 3	180 days

Conditional Upload

- By default a file is overwritten if same key is present.
- Can be avoided by conditional upload.
- Only using API or CLI
- AWS CLI version should be at least 2.19.1

```
aws s3api put-object --bucket 3aayaam-s3-bucket --key retail/orders_2_10.csv --  
if-none-match "*"
```

Object Multipart Upload



- Break the file and upload in parts.
- Parts can be uploaded in any order.
- S3 assembles and creates the file before putting in it the bucket.
- If one part fails, only that part can be retried.
- Should be used when object size is more than 100MB.

Multipart Upload Steps

- Break the file into parts
- Initiate multi-part upload.
- Upload parts in parallel.
- List parts.
- Complete upload.

```
split -b 50M -d <file-name>
```

```
aws s3api create-multipart-upload
```

upload-id

S3

```
aws s3api upload-part --upload-id
```

```
aws s3api list-parts --upload-id
```

```
aws s3api complete-multipart-upload --upload-id
```



Object Operations



- Copy
- Move
- Delete
- Rename
- Download



S3 Object Locks



- Retention period – Duration for which the object cannot be modified.
- Legal hold – Unlimited retention. Cannot be deleted unless legal hold is removed.



Object Encryption



- Bucket level encryption applies to all objects inside of it.
- If certain objects have to be encrypted with a different key then object level encryption can be used.

Hands-on

- Done

Summary S3 Basics

- Distributed Storage and S3.
- S3 Object Store. S3 in the data engineering pipeline.
- S3 storage hierarchy.
- Buckets and objects. Upload objects using console and CLI.
- Bucket properties - versioning, encryption, access logging, event notifications, transfer acceleration, requestor pays.
- S3 Storage Class.
- Conditional and multi-part uploads for objects.
- Object operation, object locks and object encryption.



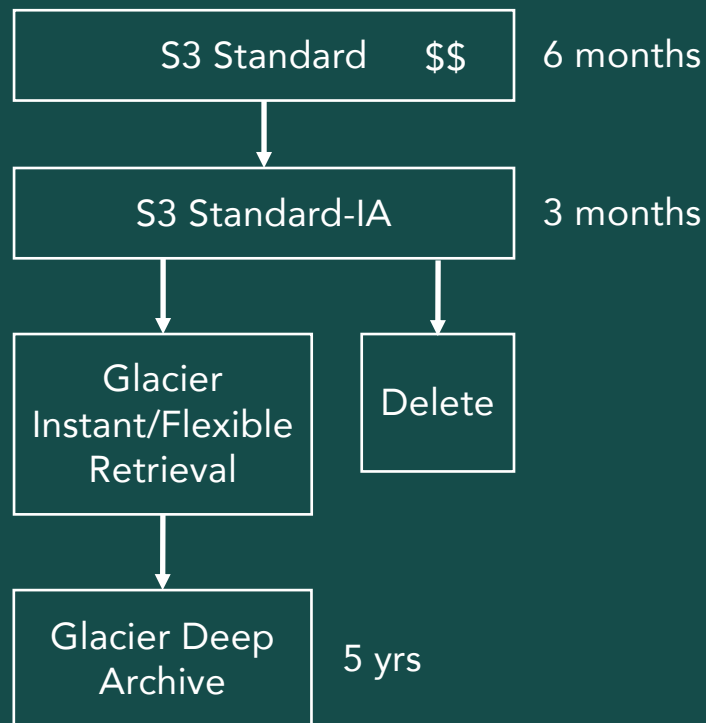
S3 Management

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Agenda

- Lifecycle Policies
- Cross Region Replication
- Bucket Metrics

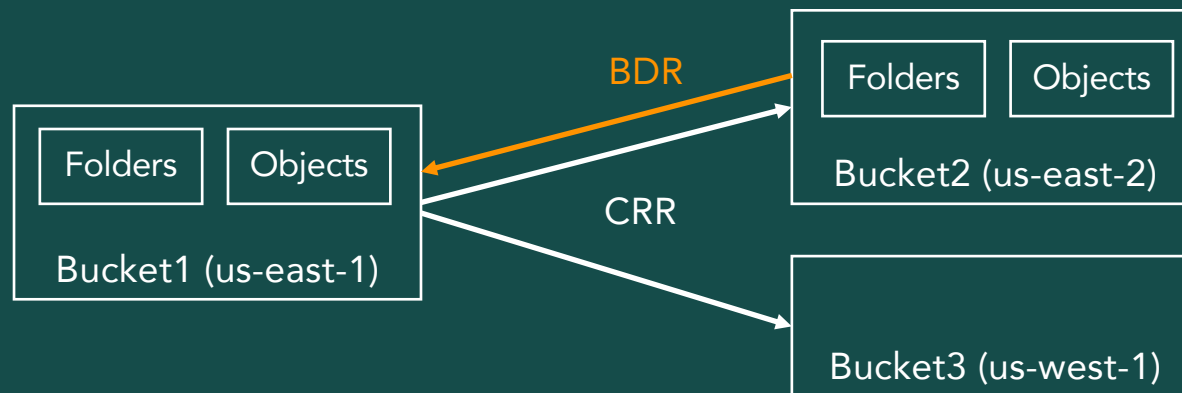
Lifecycle Policies



- Automatically move objects from one storage class to another.
- Reduce cost.
- Bucket-level, folder-level, size-wise.
- Actions:
 - Transition – Move across storage classes.
 - Expire / Delete – Delete objects.
- Current versions & Older versions.
- Versioning has to be enabled.
- Rule conflicts.

Cross Region Replication (CRR)

- Replicate objects across or within regions to a different bucket.
- Use cases:
 - Disaster recovery.
 - Global deployment.
- Replication types:
 - Existing objects (batch replication).
 - New objects (live replication).





Bucket Metrics

- Total Size of all objects.
- Total no. of objects.
- Replication metrics.





S3 Advanced Features

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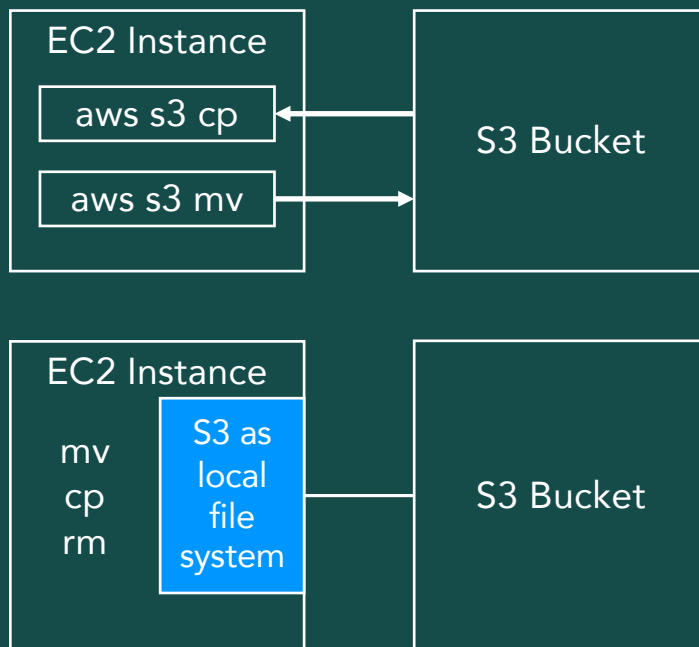
Agenda

- S3 Mountpoint
- Object Lambda
- S3 SELECT
- Express One Zone
- S3 Batch

S3 Express One Zone

- Stores data in a single AZ.
- Used for low latency use cases. Data access speed can be 10x faster.
- Directory Buckets.
- Multipart upload can be used. Steps: create-upload, upload-parts, list-parts and complete-upload.
- S3 Lifecycle policy does not support transition (move from one storage class to another) actions. Only, expire objects.
- Different IAM policy to perform S3 operations.

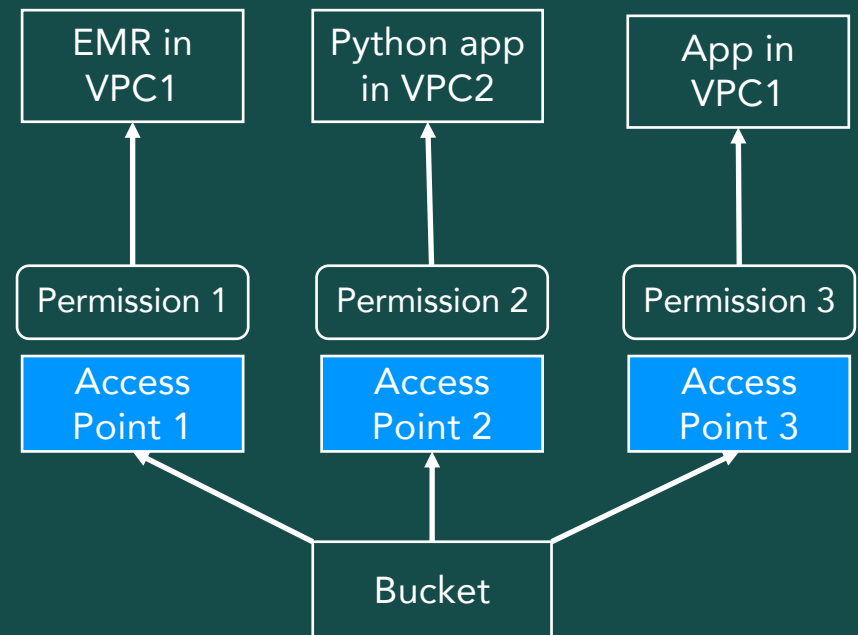
S3 Mountpoint



- S3 bucket is mounted as a local file system in EC2.
- OS calls are translated to S3 API calls automatically.
- Read heavy workloads.
- Install mountpoint.
- Mount S3 bucket as local file system in EC2 instance.
- Caching can be configured for the mountpoint.

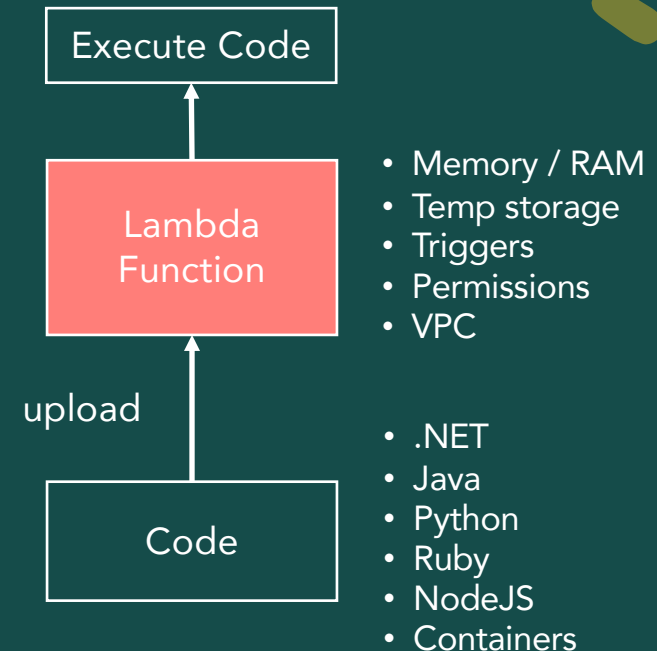
Access Points

- Provides different endpoints for the same bucket.
- Different permissions can be attached to different endpoints.
- Access through URLs only.



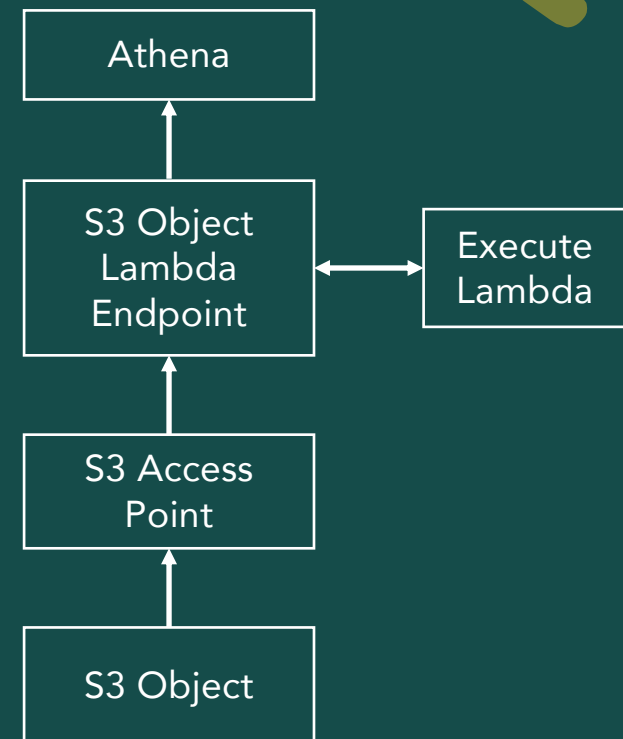
Lambda Primer

- Serverless compute.
- Compute – CPU & RAM.
- Why: To execute codes and programs.
- When: Manually or automatically based on triggers.
- Write code and upload to Lambda. It will execute based on triggers (manual or automatic).
- Lets explore the console, triggers.



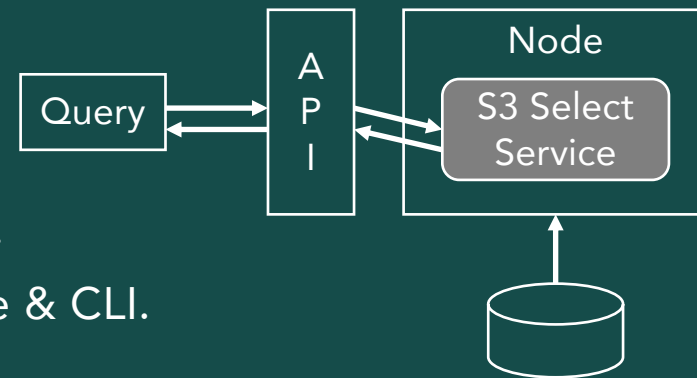
Object Lambda

- Can be used to process (program logic using code) data in files/objects before returning.
- Can be used with Athena, EMR Spark, Redshift Spectrum.
- Works at bucket level. Logic has to be written in code to handle object level processing.
- Use cases:
 - Data masking.
 - Transform data before returning to application.



S3 Select

- Filter content of object.
- Retrieve subset of data.
- Reduces amount of data that is sent from S3.
- Can query only one object at a time. Console & CLI.
- CSV, Parquet, JSON. BZIP2 & GZIP format.
- Storage classes – Standard, Standard-IA, Express One Zone, One Zone-IA, Glacier Instant Retrieval.
- SQLs – SELECT ... FROM <object> WHERE ... LIMIT ...
- Functions: avg, sum, min, max, sum, case, coalesce, cast, date& string functions.



S3 Batch

- Uses job to perform operations on large no. of objects.
- Manifest file contains list of all objects on which action has to be performed.
- Supported operations: Copy objects, Lambda function executions, Operations on tags, apply object lock retentions or object lock legal holds.
- Use case:
 - Create thumbnails of all videos or images in a folder.
 - Tagging objects in one or more folders.
 - Copying huge no. of objects from one place to another or across different storage classes.
 - Video transcoding.



S3 Storage Lens

- Analytics on S3 storage and activities.
- Organization level dashboards.
- Needs IAM user to access dashboards.

Summary

- Lifecycle Policies, Cross Region Replication.
- S3 Express One Zone.
- S3 Mountpoint.
- Access Points.
- Object Lambda.
- S3 Select.
- Batch operations.
- S3 Storage Lens.

Feature-wise use case

- Versioning – Protect against accidental object modifications.
- Encryption, Server Access Logging & Object Lock – Security.
- Transfer acceleration – Faster upload time over long distances.
- Event Notifications – Get notifications or event driven processing / execution on object operations.
- Storage Class – Different workload and pricing.
- Multipart upload – Parallel and faster object upload/copy.

Feature-wise use case

- Lifecycle Policies – Storage cost optimization, automation.
- CRR – Governance, compliance, global deployment.
- Express One Zone – Faster application access to objects.
- Mountpoint – Simplify S3 API operations using Unix FS commands.
- S3 Select – Filter object data to reduce amount of data transfer.
Can be done only one object at a time.
- Object Lambda – Execute logic before sending data to application.
Athena, Redshift Spectrum etc.



S3 Security

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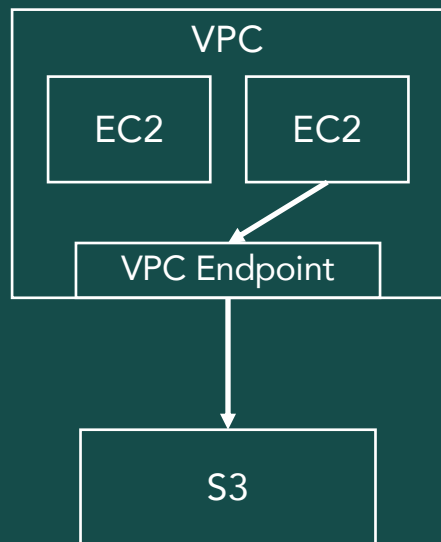
Agenda

- IAM
 - Identity based policy
 - Resource based policy for S3 (Bucket policy)
- VPC Endpoint
- Access Points
- Pre-signed URLs

IAM

- Authentication & Authorization.
- S3 authorization is done using IAM policies.
 - Identity based policy.
 - E (Effect) + A (Action) + R (Resource)
 - Resource based policy (Bucket Policies).
 - E (Effect) + A (Action) + R (Resource) + P (Principal)
- S3 actions – CreateBucket, DeleteBucket, CopyObject, DeleteObject, GetObject, PutObject, ListAllBuckets, ListBucket, Describe* etc.

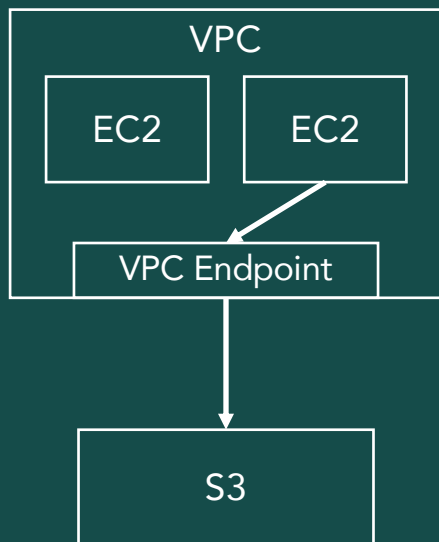
VPC Endpoint Access



- Resource based IAM policies can be used to control access from specific VPC endpoints.

```
{
  "Statement": [
    {
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::awsexamplebucket1"],
      "Condition": {
        "StringNotEquals": {
          "aws:SourceVpce": "vpce-111bbb22"
        }
      }
    }
  ]
}
```

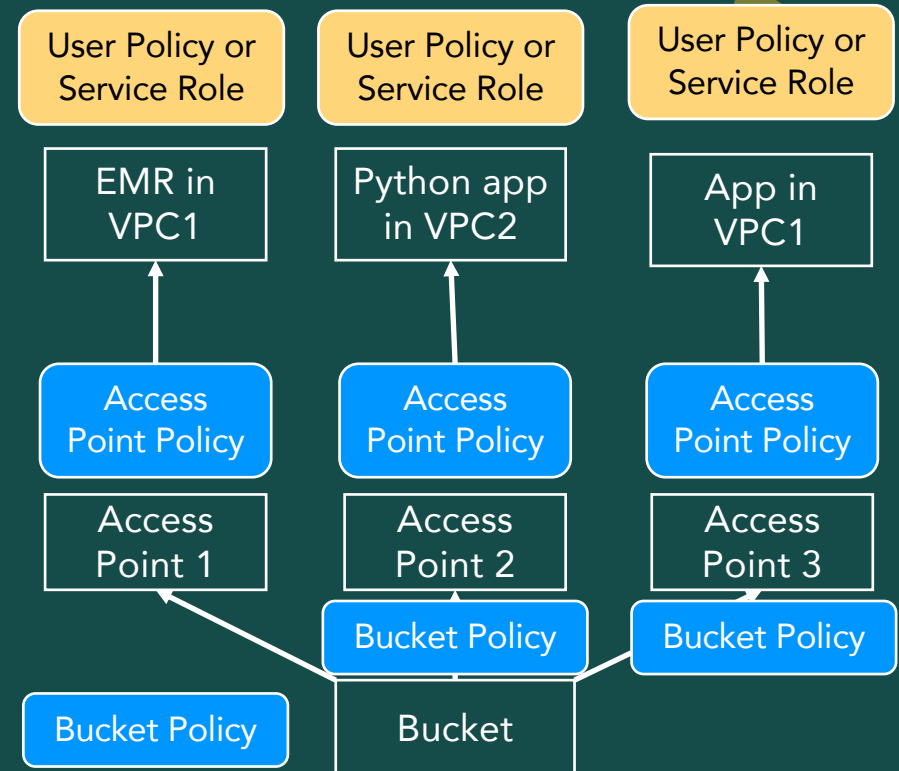
VPC IP Address Access



```
{
  "Statement": [
    {
      "Principal": "*",
      "Action": "s3:*",
      "Effect": "Deny",
      "Resource": ["arn:aws:s3:::awsexamplebucket1"],
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "192.0.2.0/24"
        },
        "NotIpAddress": {
          "aws:SourceIp": "192.0.2.188/32"
        }
      }
    }
  ]
}
```

S3 Bucket Access Points

- Individual IAM policies for individual access points.
- EAR, EARP policies.
- Resource – ARN of access point.
- IAM policy for Bucket – Resource based or bucket policy. EARP.
- IAM policy for Access Point – Resource based or bucket policy. EARP.
- IAM policy for app/user – Identity based policy. EAR.



Pre-signed URLs

- Time limited access to access objects.
- Provide temporary access to users to use S3 objects without IAM policies.
- Console or CLI.
 - `aws s3 presign s3://<bucket>/<object> --expires-in 3600`
- Can be used to download and upload objects.



Others

- Encryption – At rest using SSE and in-transit using TLS.
- ACLs (Access Control List) – Resource based option to control access to buckets and objects.

Summary

- IAM
- Restrictions to VPC Endpoint and VPC.
- Bucket access points.
- Pre-signed URLs.

Performance Considerations

- S3 scales automatically to handle concurrent requests.
- 3500 writes and 5500 reads per second per prefix (folder).
- If a bucket has 20 folders . . .
 - 70,000 writes per second.
 - 110,000 reads per second.
- EC2 instances and S3 should be in the same region.
- Transfer acceleration.

S3 Pricing (us-east-1)

- Storage

Storage Class	
Standard	First 50TB - \$0.023/GB Next 450TB - \$0.022/GB Over 500TB - \$0.021/GB
Standard-IA	\$0.01235/GB
One Zone-IA	\$0.01/GB
Express One Zone	\$0.16/GB
Glacier Instant Retrieval	\$0.004/GB
Glacier Flexible Retrieval	\$0.0036/GB
Glacier Deep Archive	\$0.00099/GB

- Example – 678 TB per month is stored in standard class and after 3 months it is moved to glacier instant retrieval for 30 days and then moved to glacier deep archive and kept there for 10 years.
- Standard storage class = $(50\text{TB} + 450\text{TB} + 178\text{TB}) = (50 * 1024 * 0.023) + (450 * 1024 * 0.022) + (178 * 1024 * 0.021) = \15143 .
- Glacier instant retrieval = $678 * 1024 * 0.004 = \$2777$.
- Glacier deep archive = $678 * 1024 * 0.00099 = \$688$.
- Total monthly cost = \$18608.

S3 Pricing (us-east-1)

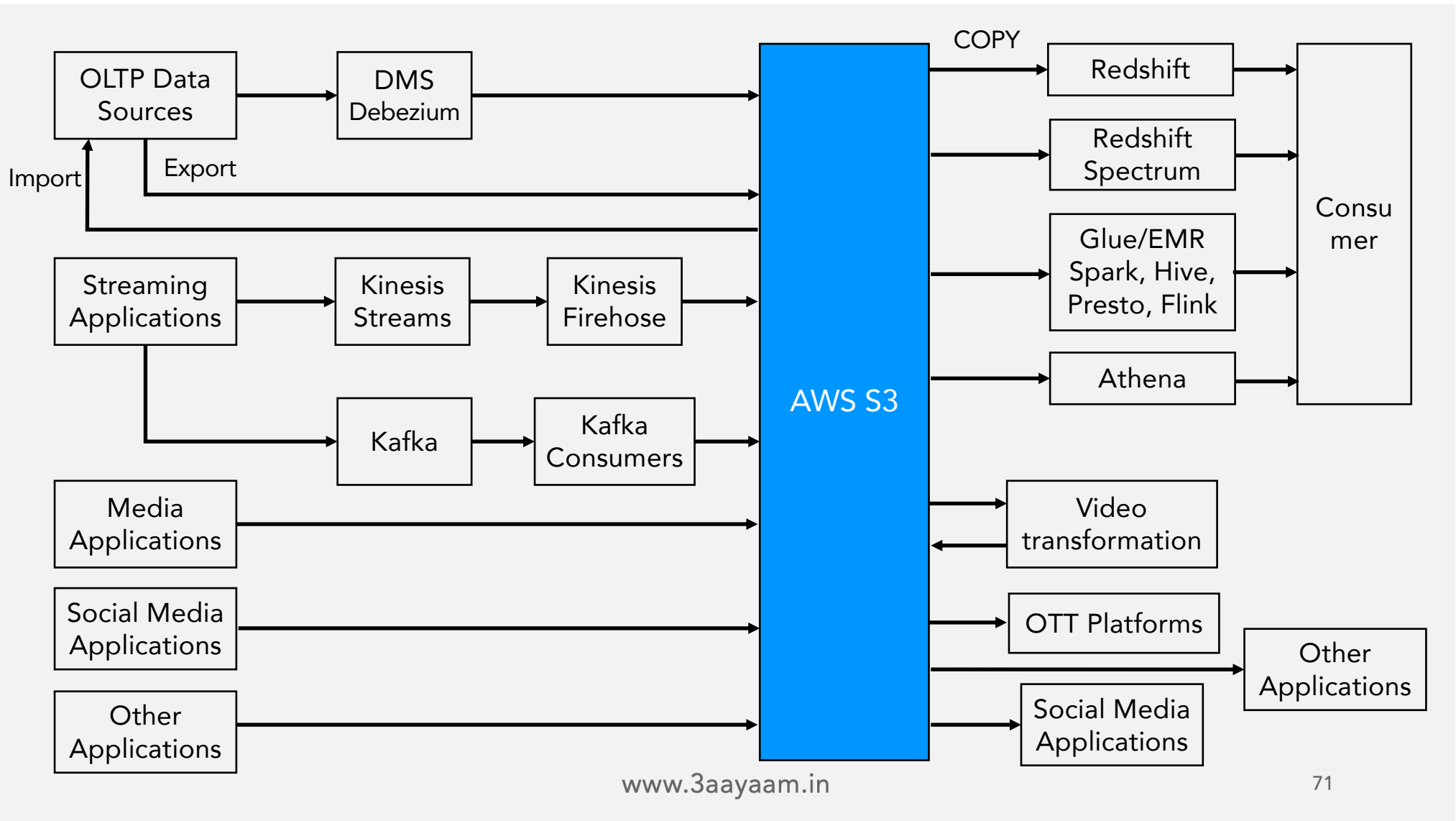
- Cost of Upload, Copy, Select, Download. \$0.005 per 1000 requests.
- Example – Records from Aurora MySQL and DynamoDB are exported to S3 using cron job that is scheduled daily. For Aurora 15 jobs run per day to upload 15TB of data and DynamoDB export (incremental export of) to S3 is used to upload 12 files a day from different tables.
- There are applications that use S3 to store user user uploaded documents. Total active users are approx. 500,000 and they upload 2-3 documents per day.
- Total PutObject per day = $15 + 12 + (500000 * 3) = 1500027$
- PutObject per month = $1500027 * 30 = 45000810$
- Total upload cost = $(45000810/1000) * 0.005 = \225.00

S3 Pricing (us-east-1)

- Encryption – Only for DSSE-KMS = \$0.003/GB/month.
- Data Transfer – \$0.02/GB.
- S3 Batch (for batch processing and batch replication)
 - Jobs = \$0.25 per job
 - Objects = \$1.00 per million objects processed
- Replication (CRR) – 1) Storage charges in target region. 2) Replication PutObjects. 3) Inter-region data transfer out.
 - Ex: 250TB of data replicated for Aurora and DynamoDB. 27 PutObjects per day.
- S3 Object Lambda – Amount of data returned after processing by Lambda. \$0.005/GB of data returned. GetObject + Lambda execution + Object Lambda

Architecture using S3

- Redshift Spectrum
- Athena
- Spark on EMR
- Glue
- Kinesis Firehose
- AWS DMS
- Presto on EMR
- HBase on EMR



Thank you

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